ASEAN GEOAI FUSION 2026

Innovator Traits • Innovation Process • Design Thinking • Stakeholder Engagement

Becoming a GeoAI Innovator

Turn ideas into real-world impact with GeoAI



WHAT YOU'LL EXPLORE

What sets successful GeoAI innovators apart
Mindset, curiosity, impact, and resilience.

The Design Thinking framework, step by step
Empathize → Define → Ideate → Prototype → Test

Building a problem-solving mindset
See problems as opportunities for meaningful change.

Hands-on Activity 2 — Design Thinking Exercise
Apply the framework to a real challenge.

THE INNOVATOR JOURNEY

Empathize
Understand people and the real-world context.

Define
Turn insights into a clear, meaningful problem.

Ideate
Generate many ideas and choose the most promising.

Prototype
Build quickly, learn early, and iterate.

Test
Validate with users and refine for impact.

WHAT YOU'LL TAKE AWAY

- A GeoAI innovator mindset
- A clear, repeatable innovation process
- Tools to solve real-world problems
- Confidence to build and test your ideas
- Skills to work with people and create impact

WHO IT'S FOR
Students, young innovators, and problem-solvers

FOCUS
People • Process • Purpose • Impact

OUTCOME
Stronger mindset. Better solutions. Real-world impact.

Day 1 • 7 July 2026

"Great GeoAI solutions start with great questions — and innovators who care enough to solve them.

Use location intelligence

Solve real problems

Create positive impact

Be the Innovator Our Future Needs

Where We're Going Today

Five parts, each **building** on the last.

1 What is GeoAI?

Plain-language definition, real examples, key terms.



2 How it gets built

The journey from problem to trusted system.



3 Build one together

Step by step: a real flood-risk dashboard, start to finish.



4 The skills involved

Technical and human skills — both matter.



5 Your learning path

How to start from where you are today.



- Understand GeoAI
- See the end-to-end development process
- Build a hands-on project together
- Develop the right mindset and skills
- Plan your next steps

Six Terms You'll Hear Today



Everything else builds on these — **keep this slide handy.**

GIS

Geographic Information System — software for storing, analyzing, and mapping location data. Think **"Google Maps for professionals."**

Remote sensing

Collecting information about the Earth from a distance — usually satellite or drone imagery.

Model

The "brain" on AI system learns. You train it on examples; it then makes predictions on new data.

Labels

The correct answers used for training — e.g., photos of fields marked **"rice"** or **"not rice"** by a human.

CRS

Coordinate Reference System — the convention that pins data to the right spot on Earth. Get it wrong and maps misalign by kilometres.

Deployment

Moving a model out of the lab into a real tool people use — an app, dashboard, or alert system.

Know the building blocks › **Understand how they work** › **See how they fit together** › **Build amazing GeoAI solutions!**

GeoAI in the Real World

Three examples you can picture — no jargon required.

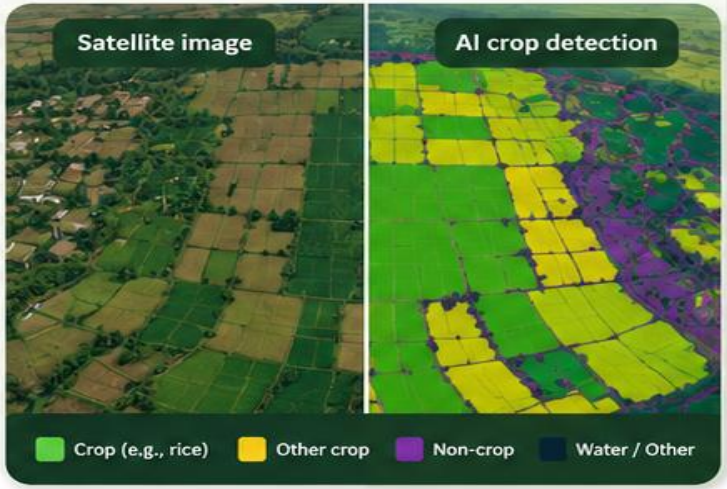
After a hurricane

AI reads fresh satellite photos to map which roads are blocked and which buildings are damaged — so rescue teams know where to go first.



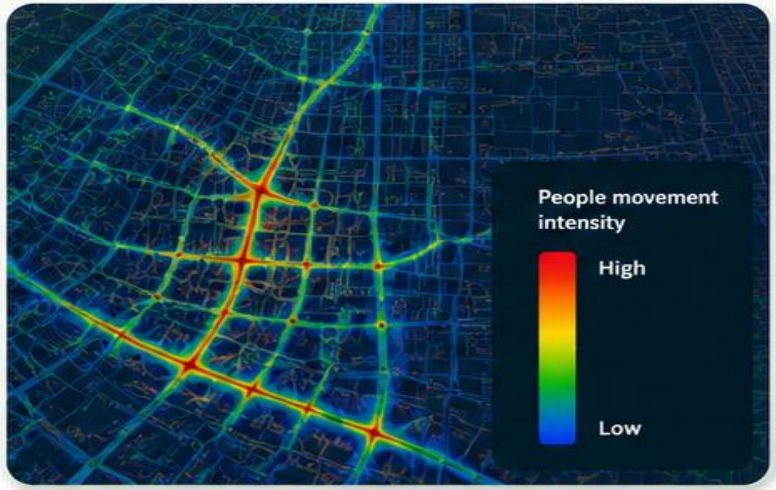
On the farm

Satellites photograph fields every few days. AI tells crop from non-crop, helping farmers and agencies make time-sensitive decisions.



In the city

AI estimates how people move around town from satellite imagery — replacing travel surveys that are too costly to run often.



What Is GeoAI?

Geospatial data

Maps, satellite images, GPS traces — anything tied to a place.



+

Artificial intelligence

Software that learns patterns from examples instead of fixed rules.



=

GeoAI

AI that understands “where” — and turns it into decisions.



In practice, GeoAI answers questions like:



Which roads are **still passable** after the flood?



Which farm fields are planted with **rice** this season — across an entire state?



Where do people **actually travel** in this city, even without expensive surveys?

Why “Where” Makes AI Harder

Location data has quirks that ordinary AI projects don’t face. **Two matter most.**

Places are all different

A model trained on one city, season, or satellite often fails in another. Researchers call this spatial heterogeneity — for us: **“what works in Penang may not work in Johor.”**



Location is sensitive

Where someone lives, travels, or works can reveal a lot about them. GeoAI carries extra privacy, fairness, and misuse risks — so **responsibility is built into the work, not added later.**

- **Privacy**
- **Fairness**
- **Misuse risk**
- **Responsibility by design**

Takeaway: GeoAI is not just “running AI on maps.”
It needs people who understand **both the technology and the geography.**

How a GeoAI Solution Gets Built

Before the details: every project moves through **four big phases** — and then loops back.

1 Frame



Understand the real problem and who needs it solved.
What decision are we improving?

2 Data



- Maps
- Images
- Labels

Collect and clean the maps, images, and labels the AI will **learn** from.

3 Model



- Train
- Test
- Validate

Choose, train, and test the AI — making sure it works in the **real world**, not just the lab.

4 Operate



Put it in users' hands, watch it, maintain it, and keep it **safe and fair**.

And back to Frame —
real projects loop as data, needs, and conditions change.

Zooming In: The Eleven Stages

The same four phases, broken into the steps professionals **actually** follow.

FRAME
Define the right problem and goals

1 Problem identification
Understand the real problem and who needs it solved.

2 Requirements analysis
Define objectives, scope, constraints, and success criteria.

DATA
Gather and prepare the right information

3 Data collection
Gather maps, satellite images, sensors, and other relevant data.

4 Data preparation
Clean, organize, and transform data into usable formats.

5 Exploratory analysis
Explore patterns, check quality, and understand the data better.

MODEL
Teach the AI to learn and make predictions

6 Model design
Choose the right approach, algorithms, and features.

7 Training & validation
Train the model, test its performance, and validate in the real world.

OPERATE
Put it to work and keep improving

8 Deployment
Deploy the model as a tool, app, or service.

9 Decision support
Deliver insights to users and support better decisions.

10 Monitoring
Track performance, detect issues, and measure impact.

11 Governance & ethics
Ensure privacy, fairness, transparency, and responsible use.

Notice: Governance and ethics is a stage of its own — being trustworthy is part of the job, not an afterthought.

Where Projects Actually Fail



Surprise: it's rarely the AI itself. The research points to **four repeat offenders**.



1 Solving the wrong problem

The tech works,
but nobody needed
that answer.
Framing was skipped.



2 Weak training labels

The AI learned from
wrong or outdated
examples — **garbage in,
garbage out.**



3 It didn't travel well

Great results in the
pilot region;
poor results
everywhere else.



4 Nobody trusts the output

No explanations,
no confidence levels —
so decision-makers
ignore it.

The lesson: frame the **decision** first, engineer the **data** second, design the **model** third — and build in **monitoring** and **governance** from day one.



PART 3

Let's Build One Together

A step-by-step mini case study: a **flood-risk dashboard** for a flood-prone district. Fictional, but modeled on real **GeoAI deployments**.



THE ASK, FROM THE DISTRICT OFFICE

> *"Every monsoon season we scramble. Can we see which neighbourhoods are most at risk — before the water arrives?"*



NINE STEPS — ENDING WITH THE FINISHED PRODUCT

1. Problem identification
2. Requirements analysis
3. Data collection
4. Data preparation
5. Exploratory analysis
6. Model design
7. Training & validation
8. Deployment & delivery
9. Monitoring & governance

1 Turn the Ask Into a Clear Problem

PHASE: FRAME | LIFECYCLE STAGE: Problem identification



Before touching any data, we interview the people who will use the answer. Three questions do most of the work:

What decision improves?

Prioritizing evacuations and drainage works — not "making a map."

Who uses it, and when?

The district disaster unit, every morning during monsoon season.

Why AI at all?

Too many interacting factors (rain, terrain, drainage) for simple rules.

PROBLEM STATEMENT (1 PAGE)

Decision: rank neighbourhoods by flood risk each morning.

Users: disaster unit (primary), town planners (secondary).

Spatial unit: neighbourhood. | **Update:** daily, 6 am.

Success: warnings hours earlier; ≥85% match with real flood events.

Out of scope: house-level predictions (privacy + accuracy).

What we produced: a one-page problem statement everyone signs before any technical work starts.



2 Agree the Rules of the Game

PHASE: FRAME | LIFECYCLE STAGE: Requirements analysis



We translate the problem into concrete, testable requirements — **so "done" means the same thing to everyone.**

Accuracy

≥85% agreement with real flood events, checked per area.

Freshness

New risk map every morning by 6 am, automatically.

Explainability

Every flagged area shows why: elevation, rain, history.

Privacy

Neighbourhood aggregates only — never household locations.

Usability

One screen; usable by officers with no GIS training.

Oversight

Officers can override or flag wrong predictions.

What we produced: a signed requirements sheet — our contract for the rest of the project.



3 Gather the Ingredients

PHASE: DATA | **LIFECYCLE STAGE:** Data collection

We pull together the right data from trusted places — each one adds a key piece of the picture.



DATA SOURCE	WHAT IT TELLS THE AI	WHERE IT COMES FROM
Satellite imagery	What the land surface looks like right now	Free public satellites (e.g., Sentinel-2)
Elevation model	Where water naturally flows and pools	National mapping agency
Rainfall records	The trigger: how much rain, how fast	Meteorological department
Past flood maps	The labels — our "correct answers" for training	District records + expert marking
Roads & drainage	Context that changes how water behaves	Local GIS office



What we produced: a data inventory — every source listed with its license and permission to use.

4 Clean, Align, and Split

PHASE: DATA | LIFECYCLE STAGE: Data preparation



Raw geographic data never lines up on its own. This unglamorous step is where most project time goes.

1 Align

Put every layer in the same coordinate system (CRS) so maps overlap correctly.



2 Repair

Fix broken shapes, fill gaps, remove cloud-covered images.



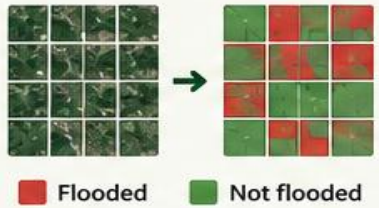
3 Tile

Cut big images into small squares the AI can digest.



4 Label

Attach the "flooded / not flooded" answers to each tile.



5 Split

Set aside whole areas for testing — by geography, not at random.



Beginner trap — data leakage: if test tiles sit right next to training tiles, the AI "whole separate areas.



What we produced: a clean, documented dataset — plus a script that can rebuild it from scratch.



5 Look Before You Model

PHASE: DATA | **LIFECYCLE STAGE:** Exploratory spatial analysis



We map and chart the data before any AI — cheap insight now prevents expensive mistakes later.

What we look for



- Where past floods cluster on the map



- How strongly low elevation predicts flooding



- Whether patterns differ between town and countryside

What we actually found



- Floods hug the **river** — but two hotspots don't (drainage failures)



- A **"flooded" hilltop** in the labels — a data-entry error, now fixed



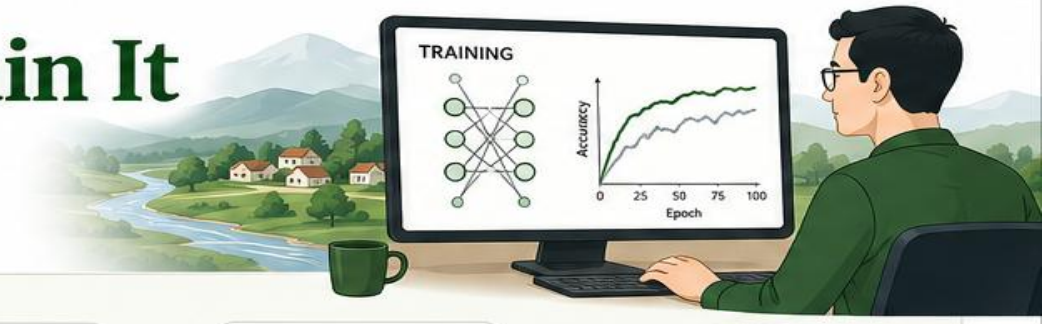
- **Rural** areas have **fewer** past **records** — we note this as a bias risk

What we produced: insight notes and corrected labels — the model will only be as honest as these.

Insight notes | Corrected labels | Exploratory charts | Maps & findings

6 Choose the Model, Then Train It

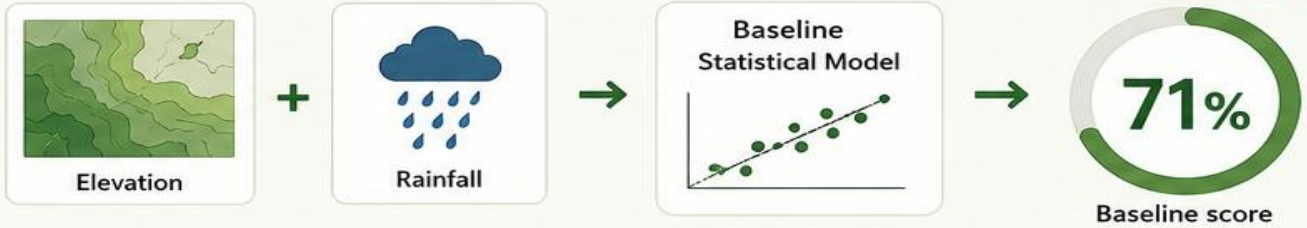
PHASE: MODEL | LIFECYCLE STAGE: Model design + training



Start simple

First a basic statistical model using elevation + rainfall.

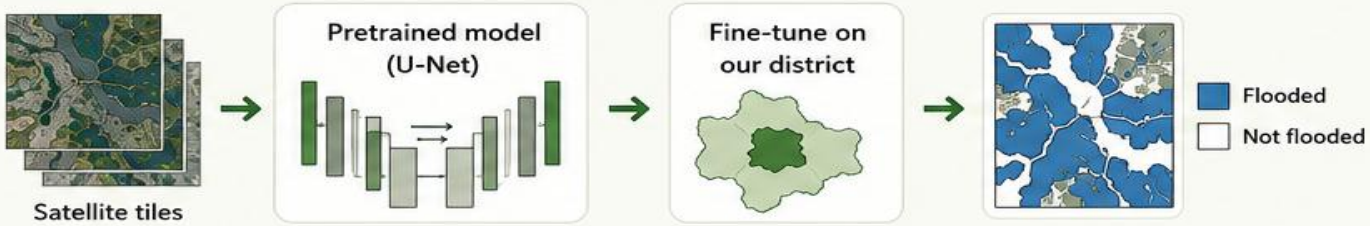
It scores **71%**. This is our baseline — the score any fancier model must beat.



Then step up

Next, a deep-learning model that "sees" satellite tiles (image segmentation).

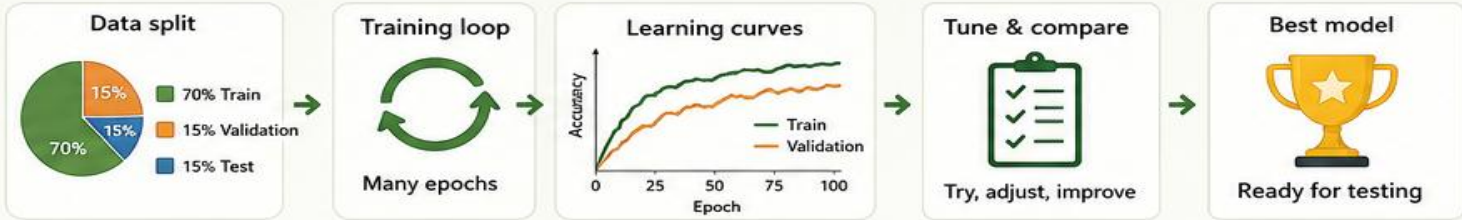
We reuse a pretrained model and fine-tune it on our district — far cheaper than training from zero.



Train & tune

The model studies tiles from 70% of neighbourhoods, guessing and correcting itself thousands of times.

We adjust settings and log every experiment.



What we produced: two candidate models and a training log — so results can be reproduced, not just remembered.

2 candidate models (baseline + deep learning)

Training log (all experiments)

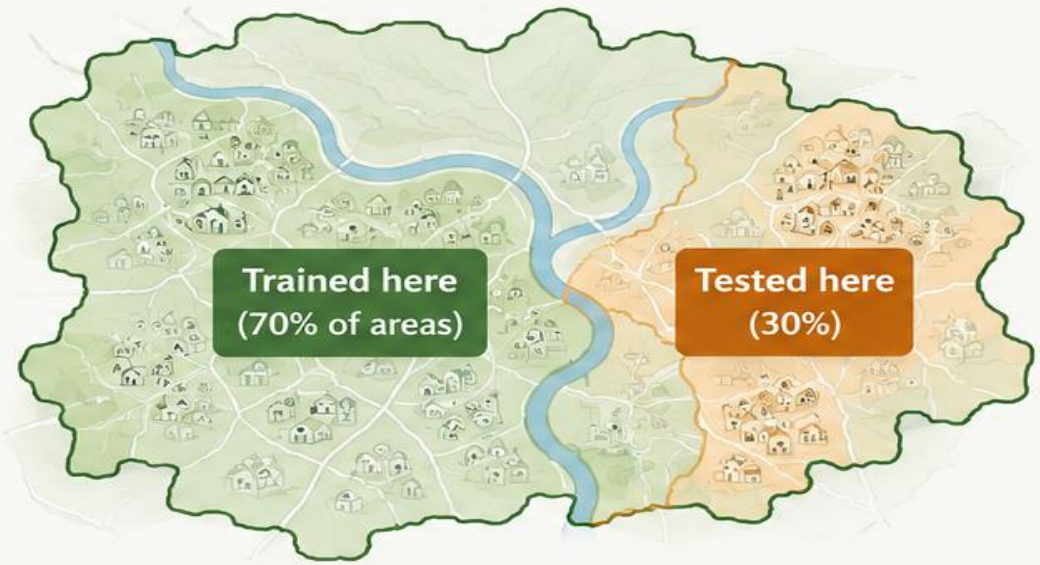
7 The Honest Exam: Unseen Areas

PHASE: MODEL | **LIFECYCLE STAGE:** Validation



We test on neighbourhoods the model has never seen — the **"spatial holdout."** This exam predicts real-world performance.

THE DISTRICT



Baseline model	**71%**
Our model — training areas	**89%**
Our model — unseen areas	**86%**

86% on unseen areas beats our ≥85% requirement — and the small drop from 89% means the model "travels" well. It also gives a confidence score per area, which the dashboard will show.

What we produced: a validation report — per-area scores, confidence, and known weak spots.



8 From Laptop to Live Service

PHASE: OPERATE | LIFECYCLE STAGE: Deployment

A model on someone's laptop helps nobody. We package it so it runs automatically, every day, the same way.



1



5:00 am

New satellite + rain data arrives

2



5:30 am

Pipeline cleans it and runs the model

3



5:55 am

Risk scores + confidence per neighbourhood

4



6:00 am

Dashboard and phone alerts update

BEHIND THE SCENES

The model is packaged in a container (a sealed box that runs identically anywhere), versioned in a registry, and can be rolled back in one step if an update misbehaves.



Containerized model

Stored in registry

Automated daily run

Monitored & versioned

One-step rollback

What we produced: a live scoring service, v1.0 — automatic, versioned, and reversible.

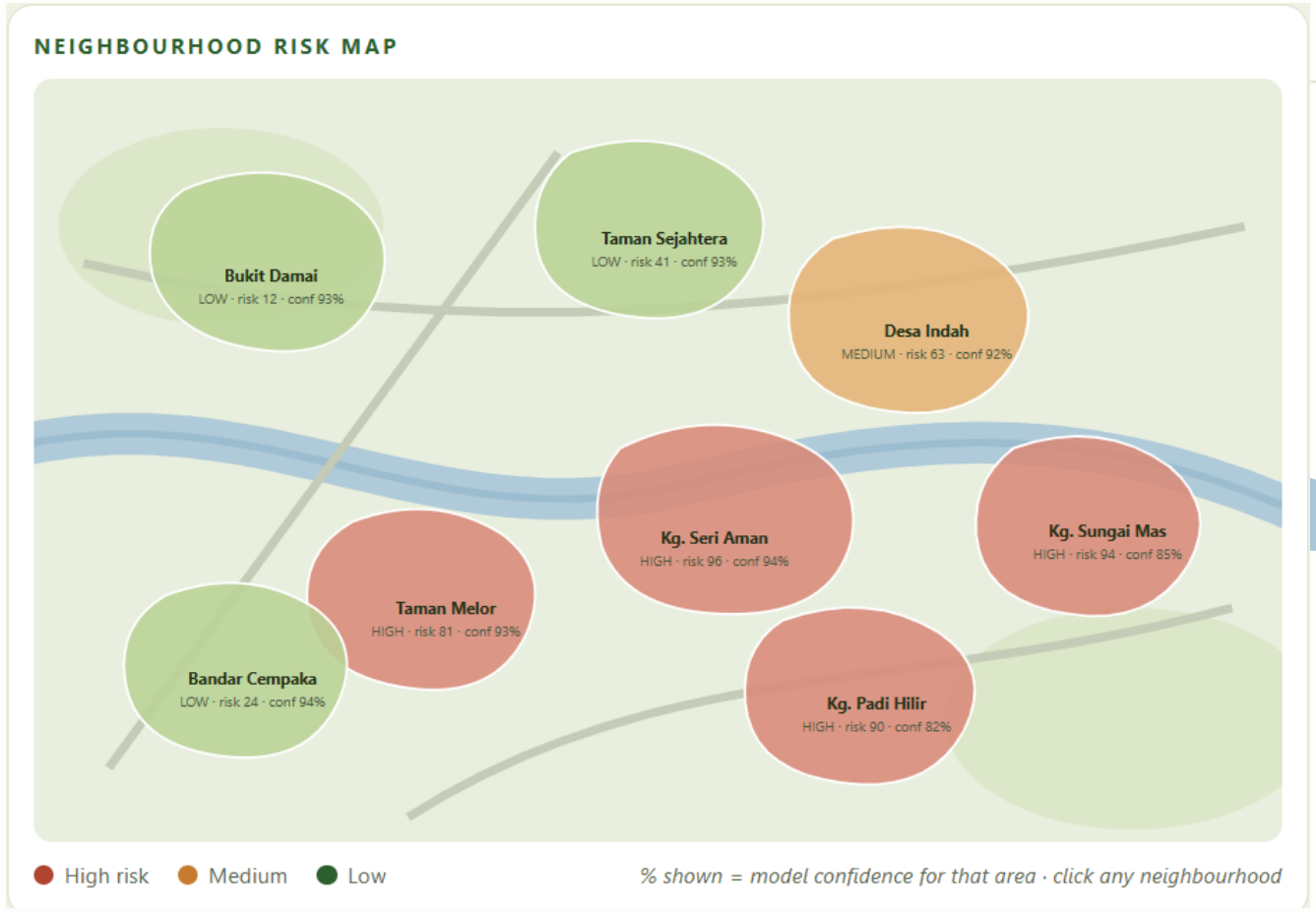
Runs automatically every day

Versioned (v1.0)

Reversible (rollback ready)

The Final Solution: A Live Flood-Risk Dashboard

What the disaster unit sees every morning at 6 am — one screen, no GIS training needed.



Model v1.0 · Updated today 6:02 am

14 neighbourhoods at high or medium risk today

≈32,400 residents in flagged zones

86% model confidence (validated on unseen areas)

ACTION QUEUE

- Kg. Seri Aman: risk rose overnight — review evacuation plan
- Jalan Tiga bridge likely cut off — reroute supplies
- Officer flagged one prediction as wrong → feeds retraining

Every element traces back to a requirement from Step 2 — accuracy, freshness, explainability, privacy, usability, oversight.

9 Keep It Accurate — and Keep It Safe

PHASE: OPERATE | **LIFECYCLE STAGE:** Monitoring + governance & ethics



Keeping it accurate

After every flood event, compare predictions to what really happened



Watch for drift — new construction and land-use change erode accuracy



Retrain each season with fresh data and officer feedback

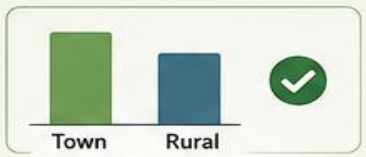


Keeping it safe

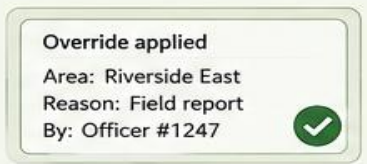
Privacy review passed: neighbourhood aggregates only, no household data



Fairness check: accuracy compared between town and rural areas



Human override: officers can correct the system — and it's logged



What we produced:

a monitoring plan and a governance scorecard — required before the district signs off.

Monitoring plan
How we track accuracy, drift, and feedback

Governance scorecard
Accuracy, fairness, privacy, override, and reliability

Ready for sign-off
Transparent, ethical, and accountable

What We Just Built, Step by Step

Nine steps, and every one produced something concrete — that's the portfolio idea you'll see in Part 5.

1. Frame → 2. Require → 3. Collect → 4. Prepare → 5. Explore → 6. Train → 7. Validate → 8. Deploy → **9. Operate**

The paper trail we produced along the way:

- ✓ Problem statement
- ✓ Requirements sheet
- ✓ Data inventory + licenses
- ✓ Clean dataset + rebuild script
- ✓ Insight notes
- ✓ Training log
- ✓ Validation report
- ✓ Live service + dashboard
- ✓ Governance scorecard

Notice what was NOT most of the work: the AI itself. Framing, data, and trust took most of the effort — exactly as the failure patterns predicted.



So What Skills Does This Take?

A common myth, corrected.

MYTH

*"A **GeoAI** expert is just a map specialist who learned AI — or an AI engineer who can read maps."*

VS.

REALITY

It's a team sport across five skill families — and the "human" skills decide whether the tech gets used at all.

1. Spatial intelligence

Thinking in maps, scale, and place.

2. Data & software engineering

Reliable pipelines, not just experiments.

4. Human & organizational skills

Framing problems, communicating, aligning people.



Great GeoAI solutions need all five.

Together, they turn data into trusted impact.

3. AI & statistics

Choosing, training, and honestly testing models.

5. Governance & responsibility

Privacy, fairness, and preventing misuse.

Better decisions
Right insights for the right places.

Trust & adoption
People trust what is fair and transparent.

Real-world impact
Solutions that are used, not just built.

Continuous learning
Better over time with feedback and data.

Accountable by design
Responsible from day one and every day.

The Technical Toolbox

Ten skill areas — nobody masters all ten. Teams combine them, and you grow into them over time.

1. Maps & GIS

Working with layers, coordinates, and spatial analysis.

2. Spatial databases

Storing and serving location data at scale.

3. Machine learning

Training and fairly evaluating predictive models.

4. Software craft

Version control, testing, and reusable code.

5. Visualization

Dashboards and maps people can act on.

6. Satellite imagery

Choosing and interpreting Earth-observation data.

7. Programming & data

Python and friends — cleaning and shaping data.

8. Deep learning

Neural networks that “see” images and read text.

9. Cloud & operations

Running models reliably in production.

10. Responsible AI

Privacy, fairness, and audit practices.

The Skills That Decide Success

Research on failed projects keeps pointing here — the human skills are **base-layer**, not optional extras.

Problem thinking

Turning a vague "we need AI" into a clear, solvable question.

Communication

Explaining results — including uncertainty — in plain decision terms.

Teamwork

Collaborating across mapping, data, engineering, and policy people.

Stakeholder skills

Balancing what users, managers, and regulators each need.

Project management

Getting from idea to delivered result, on purpose.

Design thinking

Prototyping with real users before building big.

Leadership

Driving adoption and accountability, not just launches.

Ethical judgment

Spotting potential harm early and acting on it.

Meet the GeoAI Team

Six roles that show up on real projects. You probably already resemble one of them.

GIS Analyst

The maps expert. Owns spatial data quality, analysis, and map-based outputs.

Data Scientist

The modeler. Frames prediction tasks, trains and tests the AI honestly.

ML Engineer

The builder. Turns models into reliable software and pipelines.

DevOps / MLOps

The operator. Keeps everything running, monitored, and recoverable.

Product Manager

The translator. Connects user needs to what the team builds.

Governance Lead

The conscience. Reviews privacy, fairness, and safe use before launch.

Starting From Where You Are

You don’t start from zero — you add the missing pieces to what you already know.

If you know maps (GIS)	→	Add Python programming and learn to run ready-made AI models on imagery.
If you know data science	→	Add spatial methods: coordinate systems, geostatistics, and testing across regions.
If you’re a software engineer	→	Add geospatial data formats and learn to deploy and monitor spatial models.
If you manage products or projects	→	Learn AI problem framing, success metrics, and governance checklists.
If you work in policy or ethics	→	Learn GeoAI-specific risks: location privacy, spatial bias, and misuse prevention.
If you’re brand new	→	Start with free GIS tools (QGIS) and a beginner ML course — then pick a lane above.

One Habit to Start Today: Ask the Safety Questions

Even as a beginner, you can ask these before any GeoAI system goes live. Frameworks like NIST and UNESCO formalize them.

- **Who is accountable?**
  What decision does this support, and who owns the outcome?
- **Where did the data come from?**
  Every layer and label should have a known, licensed source.
- **Could it expose people?**
  Does it reveal homes, movements, or other sensitive locations?
- **Is it fair everywhere?**
  Does accuracy drop in rural, poorer, or less-photographed areas?
- **Could it be misused?**
  Could someone repurpose it for surveillance or exclusion?
- **Can a human override it?**
  Who can pause or correct the system when it's wrong?

In Malaysia, this also means aligning with PDPA (data protection) and national AI governance guidance.

If You Remember Four Things

1

GeoAI = AI that understands “where”

It turns maps and satellite images into decisions — for disasters, farms, and cities.



2

Place makes it harder — and more sensitive

Models don't automatically transfer between regions, and location data can harm people if mishandled.



3

Success is framing + data + people, not just models

Projects fail on wrong problems, weak labels, and lost trust — rarely on the AI itself.



4

You can start from where you are

Add the missing pieces to your current role, and learn by building small real projects.



ASEAN GEOAI FUSION 2026

Responsible GeoAI

Ethics & Responsible AI for Geospatial Builders

A step-by-step guide to building GeoAI that is private, fair, accountable — and still wins the hackathon.



DO GOOD
Create real-world positive impact

DO NO HARM
Avoid causing unintended harm

BE ACCOUNTABLE
Be transparent, traceable, and auditable

RESPECT PEOPLE
Protect privacy, rights, and dignity

WHAT YOU'LL EXPLORE

People live in the places your models analyze.
Your code can impact lives.

GeoAI can amplify bias and inequality.
Be intentional. Be fair.

Trust is your advantage.
Users, judges, and communities trust responsible teams.

Responsible teams build winning solutions
that last beyond the hackathon.

A STEP-BY-STEP GUIDE TO RESPONSIBLE GEOAI

1 DEFINE THE PROBLEM WITH PURPOSE

- Understand the real need and context.
- Who benefits? Who could be affected?
- Define success beyond accuracy.

2 CHECK YOUR DATA & SOURCES

- Is the data legal and ethical to use?
- Check quality, coverage, and bias.
- Document your sources clearly.

3 PROTECT PRIVACY & RIGHTS

- Minimize data collection.
- Anonymize and aggregate where possible.
- Respect consent and local laws.

4 BUILD FAIR & ROBUST MODELS

- Test for bias across regions and groups.
- Validate with diverse data.
- Prefer explainable approaches.

5 BE TRANSPARENT & ACCOUNTABLE

- Explain how your model works (in simple terms).
- Share limitations.
- Keep an audit trail.

6 TEST IMPACT & IMPROVE

- Assess potential impacts (social, environmental, economic).
- Get feedback from users and stakeholders.
- Iterate and improve.

CORE PRINCIPLES OF RESPONSIBLE GEOAI

PRIVACY FIRST
Protect personal and sensitive location data.

FAIRNESS & INCLUSION
Design for everyone, not just the majority.

TRANSPARENCY
Be open about data, methods, and limits.

ACCOUNTABILITY
Own the impact. Fix issues quickly.

SUSTAINABILITY
Consider environmental and societal impact.

HUMAN-CENTERED
Put people, communities, and dignity first.

WHAT THIS MEANS FOR YOUR PROJECT

- Engage stakeholders early and often.
- Think beyond the demo — think real-world use.
- Design for ethics from the start.
- Measure impact, not just performance.
- Build trust. Win impact.

REMEMBER

Great GeoAI solutions combine smart technology with strong ethics.

Build solutions the world will thank you for.

WHO IT'S FOR
Students, young innovators, and problem-solvers

FOCUS
People • Process • Purpose • Impact

OUTCOME
Stronger mindset. Better solutions. Real-world impact.

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"Great GeoAI solutions start with great questions — and innovators who care enough to solve them.

Use location intelligence

Solve real problems

Create positive impact

Be the Innovator Our Future Needs

Foundations · What is spatial data?

Spatial data is any data with a location attached. It comes in three main flavours — and GeoAI models consume all of them:

1 Vector



Discrete objects with coordinates: a clinic (point), a road (line), a district (polygon) — stored as coordinate lists plus attributes.

e.g. *building footprints, road networks, admin boundaries*

2 Raster



A grid of cells, each holding a value — satellite images, elevation, rainfall, population grids. Resolution = cell size on the ground.

e.g. *Sentinel imagery, flood-extent maps, 100 m population grids*

3 Trajectories



Sequences of (location, time) points — GPS traces, vehicle movement, mobility data. The most re-identifying kind of all (see Step 3).

e.g. *phone GPS traces, ride-hailing trips, ship AIS tracks*

Every spatial record = **WHAT** happened + **WHERE** (coordinates in a reference system) + often **WHEN**. That where + when pair is exactly what makes GeoAI powerful — and what makes it risky.



Collecting vector data • a clinic map

Worked example: your team needs every health clinic in District X as points, plus district boundaries as polygons.

1 Source

Pull open data: admin boundaries from the government portal, a clinic list from the ministry or OpenStreetMap.

Example



Clinic list	
Name	District
Klinik A	X
Klinik B	X
Klinik C	X
Klinik D	X
...	...

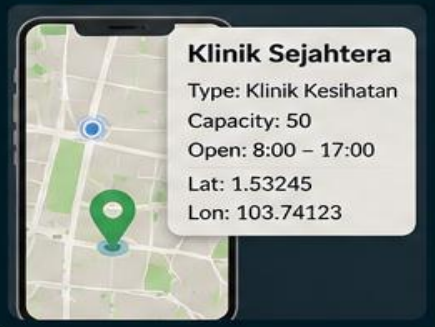
District boundaries (from portal)

Clinic list (from ministry)

2 Field capture

Verify on the ground with a GPS app — capture coordinates plus attributes: name, capacity, opening hours.

Example



3 Digitise

Trace missing buildings and access roads from aerial or satellite imagery.

Example



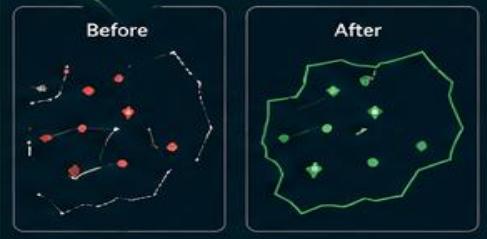
Aerial imagery

Digitised features

4 Clean

Deduplicate, fix broken geometry, align every layer to one CRS (e.g. EPSG:4326).

Example



Aligned to EPSG:4326

5 Package

Publish GeoJSON with metadata: source, collection date, accuracy, licence.

Example

```
{
  "type": "FeatureCollection",
  "name": "District X Clinics",
  "crs": "EPSG:4326",
  "features": [
    {
      "type": "Feature",
      "geometry": { "type": "Point",
      "coordinates": [103.74123, 1.53245]
      },
      "properties": {
        "name": "Klinik Sejahtera",
        "capacity": 50,
        "open_hours": "8:00-17:00"
      }
    }
  ]
}
```

clinics.geojson

ETHICS CHECKPOINTS

Consent & legality: Clinics are public facilities — but household-level points need consent and a lawful basis.

Provenance: Record provenance for every layer: who collected it, when, how, under what licence.

Fit-for-purpose accuracy: Match GPS precision to the need — don't capture centimetre accuracy for a district-level product.



Health clinic

District boundary

Collecting raster data • a flood map

Worked example: mapping flood extent after a storm using free radar imagery that sees through cloud.

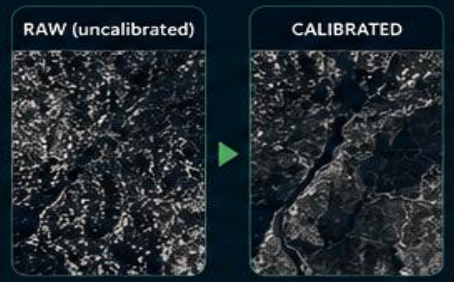
1 Acquire

The Sentinel-1 radar satellite passes over every few days; data is downlinked to ground stations.



2 Preprocess

The space agency supplies calibration and terrain correction, producing analysis-ready tiles.



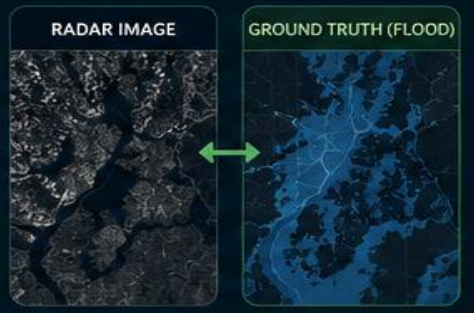
3 Download

Pull free tiles for your area and dates from the open data hub.



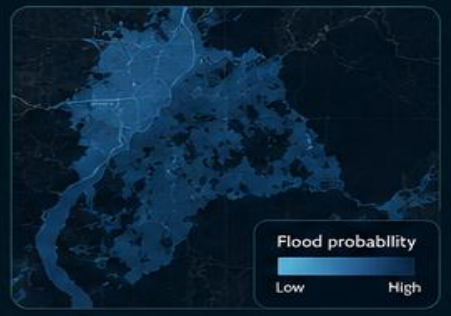
4 Label

Build ground truth: pair image patches with verified flood observations from the field or news reports.



5 Train & map

A segmentation model turns pixels into a flood extent raster — publish it with an uncertainty layer.



ETHICS CHECKPOINTS

Coverage gaps: Optical satellites are blinded by cloud; radar isn't — but revisit times still leave blind windows.

Resolution trade-off: Choose the coarsest resolution that does the job — high resolution over homes is a privacy decision, not just a technical one.

Label bias: Ask where the labels came from: if floods were only verified in accessible areas, your "ground truth" inherits that bias.



Collecting trajectory data • transit GPS

Worked example: a transit app's GPS pings become **route-improvement insights** — the most privacy-sensitive pipeline of the three.

1 Collect

The app logs a GPS ping every ~10 seconds while clear in-app notice (device, lat, lon, timestamp).



Ping example
Lat: 1.35245
Lon: 103.74123
Time: 08:15:20

2 Ingest

Encrypted pipeline: raw pings visible only to a small, role-restricted group — every access logged.



3 Clean

Filter GPS jumps and drift, then map-match pings onto the road network.



4 Minimise

Truncate trip ends near homes, reduce precision, aggregate to route segments.



5 Retain & release

Delete raw traces after a set period; publish only aggregated segment statistics.

Route segment	Avg speed (km/h)
Seg 001	21.4
Seg 002	18.7
Seg 003	24.1
Seg 004	16.9
…	…

No raw traces retained

ETHICS CHECKPOINTS

- **Consent happens at collection,** not after — the notice must say what's collected, why, and for how long.
- **Trip endpoints reveal homes and workplaces** — trim them before anyone analyses anything.
- **Vet any purchased trace data:** regulators (e.g., the US FTC) have sanctioned location brokers over unlawfully collected data.



1 Step 1 · Know What Counts as GeoAI

GeoAI = AI or machine learning applied to spatial, spatiotemporal, or Earth-observation data to **predict**, **detect**, or **decide** about things that happen somewhere.



1 Earth observation

Satellite / aerial detection, land-use maps, flood extent, vessel detection



Satellite · Aerial · Radar · Lidar

2 Place-based prediction

Neighbourhood risk scores, landslide & flood susceptibility, demand forecasts



Risk scoring · Demand · Forecasting

3 Mobility & networks

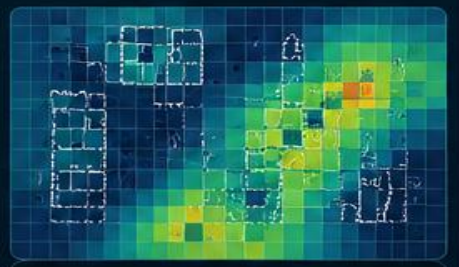
Movement modelling, traffic forecasting, route optimisation



Movement · Traffic · Routing

4 Population & settlement

Building footprints, population grids, deprivation mapping



Buildings · Population · Deprivation

5 Spatial decision support

AI outputs inside dashboards, triage, inspection & allocation tools



Dashboards · Triage · Allocation

WATCH OUT

Many hackathon projects are GeoAI without using the name.

Consent happens at collection, not after — the notice must say what's collected, why, and for how long.

Trip endpoints reveal homes and workplaces — trim them before anyone analyses anything.

Vet any purchased trace data: regulators (e.g., the US FTC) have sanctioned location brokers over unlawfully collected data.

2 Step 2 · Understand why location is different



In GeoAI, location is never “just another feature.”
One coordinate plays **three roles** at once:

1 Identity cue

Place + time can reveal who you are — home, clinic, place of worship, daily routine.

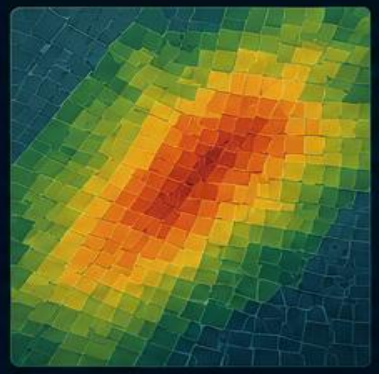


- Home 07:00
- Clinic 09:15
- Place of worship 12:30
- Market 14:45

Location + time = behavioural patterns that can identify individuals.

2 Contextual signal

Place implies attributes about the people living there — income, ethnicity, exposure.



- Income Low → High
- Ethnicity Distribution
- Exposure risk Low → High

Location = proxy for sensitive attributes, even without names.

3 Lever for action

Maps direct policing, welfare scrutiny, and disaster resources at real neighbourhoods.



Location determines where power, resources, and risk are applied.

CORE PRINCIPLE

The same map tile can describe a place, imply things about its people, and trigger action against them — so harm propagates faster than in ordinary ML.



ASEAN GEOAI FUSION 2026



MORNING BREAK

10.00 – 10.15

We resume at 10.15 with the Principles of Responsible AI

3 Step 3 · Risk #1 — Location privacy & re-identification





4 points

Just four space–time points were enough to **uniquely identify**



95%

of people in a large mobility dataset — even after coarsening.

"Anonymous" is often a myth

Pseudonymised traces stay unique; linking with outside data re-identifies people.



Location reveals the sensitive

Visits to clinics, shelters, religious sites, and homes leak health, faith & politics.



Regulators already act on this

The US FTC has sanctioned location-data brokers; California treats precise geolocation as sensitive by law.



HACKATHON RULE OF THUMB

Treat precise, linkable location data as **high-risk by default.**



4 Step 4 · Risk #2 — Spatial bias



Think of it like an exam: a random split lets the model "sit the exam" right next to questions it already saw in homework — nearby places look alike, so it just copies its neighbours. Real deployment is a brand-new exam paper.

= Training data (homework the model studied) | = Test data (the exam questions)

Random split = exam copied from homework



Test cell has training cells all around it

Every test cell has a training cell right beside it → the model copies its neighbour. Score looks great; **deployment fails.**

Spatial holdout = a genuinely new exam





A whole district the model never saw in training

Hold out a whole district the model never saw → validation now looks like real deployment. Report **per-region scores.**

WHY IT'S AN ETHICS ISSUE

- Geography correlates with wealth, ethnicity & infrastructure.
- The places your model fails are usually the places with the least data and the most need.
- Biased validation hides harm until real people are affected.
- Fairness starts with how you split the map.

4 Step 4 (deep dive) · Spatial bias — a worked example

In one sentence: the model didn't learn why floods happen — it memorised the neighbourhood.
Watch how that plays out in a real-style project:



1. Train

93%

random-split accuracy.
Looks deployment-ready.



Flood-risk model trained mostly on capital-city data — that's where the gauges and flood reports are.



Training data (mostly cities)

2. The trap

Test points sit ~200 m from training points on the same drainage network — the model learned "places near flooded" places flood."



Training points Test points (too close)

Looks great in tests.
But it's not a real exam.

3. Deploy

~60%

real accuracy in unseen rural districts.

Rolled out state-wide. Rural terrain, drainage & building patterns differ — and had almost no training data.



Low accuracy High accuracy

4. The harm



Floods missed in informal settlements with the least sensor coverage → emergency resources go to the well-mapped city, not the villages.

Delayed response

Higher risk for vulnerable communities

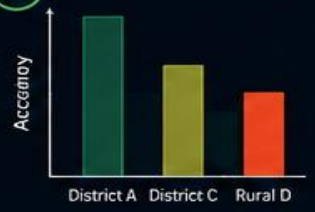
The fix (3 moves)

1. **Spatial holdout**



Validate on whole unseen districts; if accuracy drops to 68%, that's your honest number.

2. **Disaggregate**



Report metrics per district and urban/rural; one headline number always hides the weak geography.

3. **Fix the data**



Collect targeted data in under-represented regions instead of only correcting algorithmically.

5 Step 5 · Three more risks unique to GeoAI

GeoAI can harm in ways other AI systems don't.
Here are three risks to plan for from day one.



1 Surveillance & chilling effects

Outputs often drive direct action in public space — over-monitoring, false identifications, and people changing behaviour because they feel watched.

The EU AI Act now tightly restricts **real-time biometric ID** in public spaces.

2 Dual-use

The same flood map that saves lives can expose vulnerable communities; the vessel detector that protects fisheries can serve targeting.

Capability transfers even when intent doesn't — define **explicit use boundaries.**



3 Environmental footprint

Imagery pipelines, cloud inference, and constant retraining burn real energy — data-centre electricity is projected to more than double to ~945 TWh by 2030.

Right-size models and process at the **resolution you actually need.**



REMEMBER

GeoAI magnifies impact — for good or harm.
Design for trust, fairness, and sustainability.

Trust | Fairness | Accountability | Sustainability

6 Step 6 · Know the rules of the game

There is no single "GeoAI law" — compliance is assembled from **AI rules**, **privacy law**, and **geospatial sector rules**.



Three layers to check:

Global principles

OECD AI Principles
trustworthy, rights-respecting AI

UNESCO Ethics Rec.
dignity, transparency, oversight

CoE Framework Convention
first binding AI treaty:
impact assessments, remedies



AI & privacy law

EU AI Act
risk tiers; bans some public-space biometric ID

GDPR / UK GDPR
location data is usually personal data

US FTC + California
precise geolocation = sensitive; brokers sanctioned



Geospatial sector rules

INSPIRE (EU)
public spatial-data sharing standards

NOAA licensing (US)
commercial remote sensing

India Geospatial Policy
liberalised mapping, retained sensitive-feature rules



BUILDER TAKEAWAY

Assume precise location is regulated everywhere, and check **privacy law** AND **mapping rules** — they move on **separate tracks**.

Compliance | Accountability | Transparency | Global impact

7 Step 7 (deep dive) · Each tool in plain English

Start with the boring wins: collect less precision, retain less history, release **areas not points**. Then layer these tools:



Aggregation

"Report the district, not the dot."

Example: Publish "38 dengue cases in District A" instead of a pin per case.

Breaks when — Sparse areas: "1 case in a 3-house district" still points at someone's home.

k-anonymity

"Hide in a crowd of k."

Example: Generalise age 34 → 30–39, postcode 47301 → 473xx, until ≥ 5 people share every row.

Breaks when — Linkage: an attacker with outside knowledge ("she was at the mall at 9am") still finds the row. Mobility traces stay unique.

Differential privacy

"Calibrated noise = mathematical deniability."

Example: US Census style: a true count of 120 publishes as 118 — no single person's presence changes the answer noticeably.

Breaks when — Small areas & rare events: if the true count is 3, noise of ±5 destroys the signal.

Synthetic data

"Fake records, real patterns."

Example: Generate artificial trip records so the team can build & test without touching real traces.

Breaks when — Memorisation: if the generator reproduces a rare real trajectory, your "fake" data leaked a real person. Test it.

Which tool when?

- Internal dev & testing → synthetic data
- Partner data share → k-anonymity + access controls + contract
- Public dashboard / open release → aggregate, then DP on published numbers
- Always → collect less, keep less, publish areas not points

7 Step 7 · Your privacy toolbox (and its limits)



Start with the boring wins: collect **less precision**, **retain less history**, release **areas not points**.
Then layer these tools:

TOOL	BEST USED FOR	CATCH (ITS LIMITS)
Aggregation / coarser resolution	Dashboards, open releases, area-level planning	Simple & cheap — **but sparse areas** can still leak
k-anonymity	Tabular extracts, low-complexity releases	Risk reduction, **NOT proof** — weak vs linkage & mobility traces
Differential privacy	Published counts, map layers, benchmarks	Formal guarantee — **utility drops for** small areas / rare events
Synthetic data	Dev, testing, sandboxes, collaboration	Not an automatic safe harbour — **can preserve bias & leak patterns**

GOLDEN RULE: anonymisation is never one transformation — **match the tool** to the release, then **test re-identification risk** empirically.

MATCH → TEST → REDUCE RISK

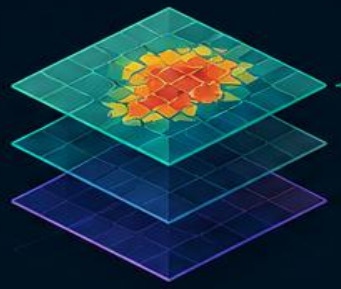
8 Step 8 · Build fairness & transparency in

Spatially-aware fairness



- **Audit** training data for geographic representation gaps
- **Report** metrics by region type, urban/rural, socioeconomic context
- **Collect** targeted new data for blind spots — don't only correct algorithmically

Spatial explainability



High risk area

Top drivers

Rainfall intensity	38%
Drainage capacity	27%
Land elevation	19%
Population density	16%

Uncertainty: High ⚠

- **Global** variable importance + **local** explanations for flagged areas
- Put **uncertainty** ON the map, not in a footnote
- Document where the model **transfers** — and where it doesn't

Documentation



- **Model cards:** intended use, performance, limits
- **Datasheets:** how data was collected & labelled
- Add **spatial metadata:** CRS, coverage, resolution, split method, weak geographies

Remember: *any single global accuracy number is hiding spatial inequality somewhere.*

Step 9 · Govern the whole lifecycle

Responsible GeoAI is a governance problem before it's a modelling problem.
Ethics starts at problem framing — not as a privacy notice at the end.

1
Frame
Public purpose & legal basis

2
Assess
Risk screen + impact assessment (DPIA / AIA)

3
Control
Technical mitigations & documentation

4
Validate
Spatial holdout + red-teaming

5
Deploy
Human oversight, overrides & appeals

6
Monitor
Audit, incident response, rollback

Human oversight must be concrete: who overrides, who signs off, how appeals work, and overrides are logged — not hidden.

Define incidents in advance (privacy leak, false hotspot, misuse) — if you can't safely stop or roll back the system, you're not ready to deploy it.

Step 10 · Learn from real deployments

LAPD predictive policing (PredPol / LASER)

Historical enforcement data treated as neutral ground truth created feedback loops that kept sending police to the same places.

Lesson: *test for feedback loops and spatial fairness before deploying.*

Sidewalk Toronto smart city

Technically sophisticated urban sensing collapsed politically over unresolved data governance, trust, and privacy questions — terminated in 2020.

Lesson: *legitimacy and governance decide success, not model novelty.*

Global Fishing Watch vessel detection

Powerful public-interest transparency at sea — but with acknowledged gaps: "dark" vessels, identity-matching limits, uneven coverage.

Lesson: *disclose dual-use, undercoverage, and interpretation limits openly.*

Global population grids

Widely used for disaster response and planning, yet found to systematically under-count rural populations.

Lesson: *audit bias in sparse areas or you will misallocate real resources.*

Common pattern: *GeoAI fails not because "the model was AI", but because place-based outputs were treated as neutral and complete when they were partial, uncertain, and socially loaded.*

Case study 1 · LAPD predictive policing

PLACE-BASED PREDICTION · POLICING

THE SYSTEM

Two programmes in Los Angeles: PredPol forecast crime "hotspot" boxes (~500 x 500 ft) from historical crime reports and directed patrols to them; LASER scored "chronic" locations and individuals using data from stops and arrests.

WHAT HAPPENED

2010s — Deployed operationally across divisions; treated recorded crime data as ground truth for where crime happens.

2019 — LAPD Inspector General audit found dataset inconsistencies, inconsistent selection criteria and weak reporting — effectiveness couldn't be demonstrated. LASER was discontinued.

2020 — LAPD ended its use of PredPol. Academic work (Ensign et al.) had formalised the "runaway feedback loop" problem.

Legacy — Now the standard cautionary example for using enforcement data as neutral truth in place-based prediction.

WHY — THE MECHANISM

Recorded incidents measure where police look, not where crime is. The loop: historical arrest data → model flags a hotspot → more patrols there → more incidents recorded there → data "confirms" the hotspot → model doubles down. Districts already over-policed get more policing regardless of true crime rates — and the model's accuracy metrics look fine throughout.

LESSONS FOR YOUR PROJECT

- Historical enforcement data is biased ground truth — ask what your labels actually measure.
- Simulate the feedback loop before deploying: what happens when your output changes the data you collect next?
- Build in independent audit and public reporting from day one, not after criticism.

Case study 2 · Sidewalk Toronto (Quayside)

URBAN SENSING · SMART CITY

THE SYSTEM

Alphabet's Sidewalk Labs partnered with Waterfront Toronto to build a sensor-rich waterfront district: mobility, energy, waste and environment sensing embedded in public space, with a proposed "urban data trust" to govern the data collected.

WHAT HAPPENED

2017 Partnership announced with an ambitious, technically sophisticated vision for data-driven urbanism.

2018 Escalating disputes over who owns and governs data collected in public space; a leading privacy advisor resigned from the project.

2019 Advisory processes documented major unresolved questions on data governance, public trust and role boundaries between the private vendor and the public agency.

2020 Development agreement terminated in May — the district was never built.

WHY — THE MECHANISM

Consent breaks down in public space: you cannot opt out of a neighbourhood. The proposed governance invented new structures (the data trust) instead of anchoring in existing public accountability, leaving role boundaries between company and city unsettled.
The trust deficit grew faster than the technology matured — legitimacy, not capability, was the binding constraint.

LESSONS FOR YOUR PROJECT

- Governance and legitimacy decide success — a technically excellent system can still fail politically.
- Engage the public before the design freezes, not after complaints begin.
- Put data ownership, purpose limits and role boundaries in writing at the start.

Case study 3 · Global Fishing Watch

EARTH OBSERVATION · MARITIME

THE SYSTEM

An open platform combining ship AIS transponder signals, satellite radar and optical imagery, and deep learning to detect and map industrial vessel activity across the world's oceans — published for anyone to use.

WHAT HAPPENED

2016 — Platform launched to make industrial fishing visible at global scale.

2024 — A Nature study (Paolo et al.) mapped industrial activity at sea and found roughly three-quarters of industrial fishing vessels absent from public tracking systems — the "dark fleets".

Today — Used for fisheries enforcement, marine protected areas and transparency research — while openly documenting its own gaps in vessel identity matching and coverage.

WHY — THE MECHANISM

This is the dual-use and undercoverage case, and a largely positive one. The same detection layer that empowers conservation can inform enforcement operations, territorial disputes or targeting. And absence of signal is not absence of vessels: AIS can be switched off or manipulated, satellite coverage is uneven, and identity matching is imperfect — so a naive reading of the map misleads.

LESSONS FOR YOUR PROJECT

- Publish known limits with the map: coverage, confidence and what absence of data means.
- Define explicit use boundaries and release tiers for sensitive outputs before someone else repurposes them.
- Transparency and humility can coexist — disclosing gaps builds credibility rather than undermining it.

Case study 4 · Global population grids

POPULATION ESTIMATION · HUMANITARIAN

THE SYSTEM

Gridded population products estimate how many people live in each map cell by combining census data, satellite building detection and ML disaggrepation. They are used operationally for disaster response, vaccination campaigns and infrastructure planning.

WHAT HAPPENED

2023

Research (Sekara et al.) found large inconsistencies between major AI-derived settlement datasets across Africa — the products disagree about where people live.

2025

A study (Läng-Ritter et al.) found systematic underestimation of rural populations across major global grids.

Today

The same products remain embedded in humanitarian and development pipelines worldwide.

WHY — THE MECHANISM

Errors concentrate exactly where alternative data is scarcest. Rural ground truth is sparse, small or dispersed buildings are missed by detectors, and census disaggrepation pulls population toward well-mapped settlements. The result is a quiet compounding bias: the communities least visible in the data receive the least accurate estimates — and therefore risk receiving the least aid, vaccines and infrastructure.

LESSONS FOR YOUR PROJECT

- Audit sparse-area and rural bias before any operational use — a global accuracy number will not reveal it.
- Cross-check multiple products and treat their disagreement as an uncertainty signal.
- Remember the stakes: resource allocation from a biased map is quiet injustice at scale.

The Ethical & Responsible GeoAI Checklist

Work through all four columns of questions before you demo. If a box stays unticked, that's your next task — not a footnote.

Privacy & Data

- [ ] Lawful basis & clear purpose defined before collecting
- [ ] Precision minimised; raw location history retention limited
- [ ] Re-identification & linkage risk tested before any release
- [ ] Access to raw coordinates role-restricted and logged

Fairness & Validation

- [ ] Training data audited for geographic representation gaps
- [ ] Spatial holdout validation on whole unseen districts
- [ ] Metrics disaggregated by region and urban / rural
- [ ] Uncertainty shown on the map, not buried in a footnote

Transparency & Accountability

- [ ] Model card + datasheet with spatial metadata published
- [ ] Intended use and known weak geographies stated
- [ ] Data provenance documented — no opaque location brokers
- [ ] Reproducible: code, config and data lineage captured

Oversight & Safety

- [ ] Human override and appeal path defined before launch
- [ ] Red-teamed for surveillance, dual-use and gaming
- [ ] Incident definitions and rollback plan ready
- [ ] Compute & energy footprint weighed against public value

Judge's-eye tip: teams that can show this checklist ticked — with evidence — stand out far more than teams with a slightly higher accuracy score.

MID-DAY BREAK · 10.45 – 11.00

Thank You

Questions & Open Discussion

Next up after the break: Team Formation & Problem Scoping for your GeoAI Hackathon project.



TRAINERS

- Assoc. Prof. Ts Dr Saraswathy Shamini Gunasekaran — Taylor's University
- Emeritus Prof. Dato' Dr Halimah Badioze Zaman — UNITEN

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